

**School of Computer Science and Artificial
Intelligence**

WEEK-10 Assignment # 20

Program : B. Tech (CSE)
Specialization :AIML
Course Title : Competitive Programming
Course Code : 23CS302PC305
Semester : VI
Academic Session : 2025-2026
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Enrollment No. : 2303A52126
Batch No. : 33
Date :25/03/26

**Wednesday Disaster Zone — Polygon Validity and PIP
Queries**

Context

During a disaster drill, responders must determine if a given evacuation boundary is valid (simple polygon) and answer quick point-in-polygon queries to decide if a location is inside the safe zone.

Tasks

1. Polygon validity:

☑ Determine if the given sequence of vertices forms a simple polygon (no self-intersection).

2. Point-in-polygon queries:

☑ For multiple query points, report whether each is inside, outside, or on the boundary.

Input format:

☑ First line: V (number of vertices)

☑ Next V lines: $x\ y$ (polygon vertices in order)

☑ Next line: Q (number of queries)

☑ Next Q lines: $x\ y$ (query points) Output format:

☑ Line 1: "VALID" or "INVALID" for the polygon

☑ Next Q lines: one of "inside", "outside", or "boundary"

Edge cases

☑ Polygon with a self-intersection

☑ Points exactly on edges or vertices

☑ Non-integer coordinates (if you extend to float)

Sample

Input:

5

0 0

4 0

4 4

0 4

2 2

3

2 2

4 2

5 5

Output:

VALID inside

boundary

outside

PYTHON:


```
File Edit Selection View Go ... Week10
Main.py x MillerRabin.java 1 MillerRabin 2
Assignment20 > Main.py > ...
51 def point_in_polygon(point, poly):
52     if orientation(p1, point, p2) == 0 and on_segment(p1, point, p2):
53         return "boundary"
54
55     # ray casting
56     if ((p1[1] > y) != (p2[1] > y)):
57         xinters = (y - p1[1]) * (p2[0] - p1[0]) / (p2[1] - p1[1]) + p1[0]
58         if x < xinters:
59             inside = not inside
60
61     return "inside" if inside else "outside"
62
63 # ----- MAIN -----
64
65 V = int(input())
66 polygon = [tuple(map(int, input().split())) for _ in range(V)]
67
68 Q = int(input())
69 queries = [tuple(map(int, input().split())) for _ in range(Q)]
70
71 # Check validity
72 if is_simple_polygon(polygon):
73     print("VALID")
74 else:
75     print("INVALID")
76
77 # Process queries
78 for pt in queries:
79     print(point_in_polygon(pt, polygon))
80
81
82
83
84
85
86
87
88
Ln 88, Col 41 Spaces: 4 UTF-8 CRLF Python 3.11.9 (Microsoft Store) Go Live
nment20/Main.py"
5
0 0
4 0
4 4
0 4
2 2
4 0
4 4
0 4
2 2
0 4
2 2
2 2
3
2 2
4 2
5 5
VALID
boundary
boundary
outside
```

JAVA:

```
File Edit Selection View Go ... Week10
Welcome Main.py PolygonCheck.java 4 x MillerRabin 2 v
Assignment20 > J PolygonCheck.java > Language Support for Java(TM) by Red Hat > PolygonCheck
1 package Assignment20;
2
3 import java.util.*;
4
5 class Point {
6     int x, y;
7     Point(int x, int y) { this.x = x; this.y = y; }
8 }
9
10 public class PolygonCheck {
11
12     static int orientation(Point p, Point q, Point r) {
13         long val = (long)(q.y - p.y) * (r.x - q.x) -
14                 (long)(q.x - p.x) * (r.y - q.y);
15
16         if (val == 0) return 0;
17         return (val > 0) ? 1 : 2;
18     }
19
20     static boolean onSegment(Point p, Point q, Point r) {
21         return q.x >= Math.min(p.x, r.x) && q.x <= Math.max(p.x, r.x) &&
22                q.y >= Math.min(p.y, r.y) && q.y <= Math.max(p.y, r.y);
23     }
24
25     static boolean intersect(Point p1, Point q1, Point p2, Point q2) {
26         int o1 = orientation(p1, q1, p2);
27         int o2 = orientation(p1, q1, q2);
28         int o3 = orientation(p2, q2, p1);
29         int o4 = orientation(p2, q2, q1);
30
31         if (o1 != o2 && o3 != o4) return true;
32     }
33 }
```

```
File Edit Selection View Go ... Week10
Welcome Main.py PolygonCheck.java 4 x MillerRabin 2 v
Assignment20 > J PolygonCheck.java > Language Support for Java(TM) by Red Hat > PolygonCheck
10 public class PolygonCheck {
25     static boolean intersect(Point p1, Point q1, Point p2, Point q2) {
28         int o3 = orientation(p2, q2, p1);
29         int o4 = orientation(p2, q2, q1);
30
31         if (o1 != o2 && o3 != o4) return true;
32
33         if (o1 == 0 && onSegment(p1, p2, q1)) return true;
34         if (o2 == 0 && onSegment(p1, q2, q1)) return true;
35         if (o3 == 0 && onSegment(p2, p1, q2)) return true;
36         if (o4 == 0 && onSegment(p2, q1, q2)) return true;
37
38         return false;
39     }
40
41     static boolean isSimple(List<Point> poly) {
42         int n = poly.size();
43
44         for (int i = 0; i < n; i++) {
45             for (int j = i + 1; j < n; j++) {
46
47                 if (Math.abs(i - j) <= 1 || (i == 0 && j == n - 1))
48                     continue;
49
50                 Point p1 = poly.get(i);
51                 Point q1 = poly.get((i + 1) % n);
52                 Point p2 = poly.get(j);
53                 Point q2 = poly.get((j + 1) % n);
54
55                 if (intersect(p1, q1, p2, q2))
56                     return false;
57             }
58         }
59
60         return true;
61     }
62 }
```

```
File Edit Selection View Go ... Week10
Welcome Main.py PolygonCheck.java 4 x MillerRabin 2 v
Assignment20 > J PolygonCheck.java > Language Support for Java(TM) by Red Hat > PolygonCheck
10 public class PolygonCheck {
41     static boolean isSimple(List<Point> poly) {
56         return false;
57     }
58 }
59     return true;
60 }
61
62     static String pointInPolygon(Point pt, List<Point> poly) {
63         int n = poly.size();
64         boolean inside = false;
65
66         for (int i = 0; i < n; i++) {
67             Point p1 = poly.get(i);
68             Point p2 = poly.get((i + 1) % n);
69
70             if (orientation(p1, p2) == 0 && onSegment(p1, pt, p2))
71                 return "boundary";
72
73             if ((p1.y > pt.y) != (p2.y > pt.y)) {
74                 double xinters = (double)(pt.y - p1.y) * (p2.x - p1.x) /
75                     (p2.y - p1.y) + p1.x;
76
77                 if (pt.x < xinters)
78                     inside = !inside;
79             }
80         }
81
82         return inside ? "inside" : "outside";
83     }
84 }
Ln 106, Col 2 Spaces: 4 UTF-8 CRLF Java Go Live
```

```
File Edit Selection View Go ... Week10
Welcome Main.py PolygonCheck.java 4 x MillerRabin 2 v
Assignment20 > J PolygonCheck.java > Language Support for Java(TM) by Red Hat > PolygonCheck
10 public class PolygonCheck {
62     static String pointInPolygon(Point pt, List<Point> poly) {
82         return inside ? "inside" : "outside";
83     }
84 }
Run main | Debug main | Run | Debug
85 public static void main(String[] args) {
86     Scanner sc = new Scanner(System.in);
87
88     int V = sc.nextInt();
89     List<Point> polygon = new ArrayList<>();
90
91     for (int i = 0; i < V; i++)
92         polygon.add(new Point(sc.nextInt(), sc.nextInt()));
93
94     int Q = sc.nextInt();
95
96     if (isSimple(polygon))
97         System.out.println("VALID");
98     else
99         System.out.println("INVALID");
100
101     for (int i = 0; i < Q; i++) {
102         Point pt = new Point(sc.nextInt(), sc.nextInt());
103         System.out.println(pointInPolygon(pt, polygon));
104     }
105 }
106
107
Ln 106, Col 2 Spaces: 4 UTF-8 CRLF Java Go Live
```

```
4
0 0
6 0
6 6
0 6
5
VALID
3 3
inside
6 3
boundary
0 0
boundary
7 7
outside
0 3
boundary
```